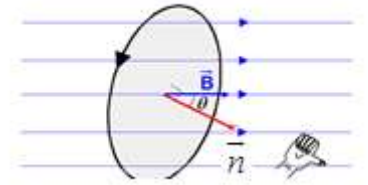


Chapter 8: Electromagnetic Induction

A- Definition of electromagnetic induction

It is the appearance of an electric current in a closed circuit without a generator by variation of the magnetic flux.



B- Magnetic flux: the magnetic flux φ of a magnetic field $\vec{B}$ through a closed surface S is given by:

$$\varphi = \vec{B} \cdot \vec{S} = B \cdot S \cdot \cos(\vec{B}; \vec{n})$$

Where:

- $\vec{n}$: is a unit vector normal to the surface S oriented in the positive direction
- B : magnetic field strength in Tesla (T)
- S : area of the closed circuit or of a closed turn in m^2

The SI unit of the magnetic flux φ is the weber (Wb).

For N turns, the magnetic flux φ becomes: $\varphi = N \cdot B \cdot S \cdot \cos(\vec{B}; \vec{n})$

C- Faraday's law:

The *change in magnetic flux* in the circuit *causes an induced electromotive* force to appear:

$$e = -\frac{d\varphi}{dt} \quad \text{or} \quad e = -\frac{\Delta\varphi}{\Delta t}$$

Hence the appearance, in the circuit of a fictitious generator which produces the induced electric current.

e is expressed in (V) and $\frac{d\varphi}{dt}$ in Wb/s.

D- Lenz law:

By its electromagnetic effects, the direction of the induced current opposes the cause that gave it rise.

E- Pouillet's law:

The current induced in the circuit is given by: $i = \frac{e}{\sum R}$ (i and e have same sign)

- i is expressed in amps (A)
- e is expressed in volts (V)
- $\sum R$ is expressed in ohms (Ω)

Note : Area of a circular turn: $S = \pi \cdot R^2$

Area of a square turn: $S = \text{side} \times \text{side}$

Area of a rectangular turn: $S = \text{length} \times \text{width}$